

I · SETUP · A ROOM, ONE RUN · 0.2

Walk one run.

A tiny consensus experiment, then the notary.
See the official record. Read the card.

`docs/demo/experiment.py` · companion to the intro deck



I • SETUP • THIS ROOM

What you will see, in order.

THE EXPERIMENT

- Twelve voters. Four candidates. Seed 0. Stdlib only.
- Aggregates and a winner are official. Individual choices are not.

THE NOTARY

- A profile that refuses to hash ballots.
- A claim card a stranger can read.
- The same seed, a new folder: a scoped match.
- A packaging bug labeled **not a science failure**.

Live path: the notebook. Talk-only path: these slides already hold the numbers.

A passing card is not a verified result.

NONE	This room uses the dry-run backend. No calendar, no pin, no gas.	Not a failed experiment.
L0	These bytes existed by time T; Bitcoin headers not verified by this tool.	Dry-run cannot earn this. Pending OpenTimestamps is not L0 either.
L1	Re-execution specified: command, git SHA, environment fingerprint.	Not a completed re-run. The command was not run by verify.
L2	Seed not grindable after the plan is anchored.	Not scientific correctness. Not today’s demo.
L3	A Sigstore keyless signature is on the reproduction receipt. Identity is recorded, not proven independent.	Not independently reproduced.

Twelve agents vote. The notary never sees a ballot.

SEED 0 · 12 VOTERS

west

north 1 · east 2 · south 4 · west 5. Shares from `summary.csv`.

A notarized winner can still be the wrong science.

WHY THIS TOY

- Same profile as the AgentFarm consensus experiment: `consensus`.
- Official artifacts are aggregates, config, report, figure.
- Ballots go under `private/` so a glob cannot admit them.
- `stdlib`. Runs in the notebook or `docs/demo/experiment.py`.

Write the folder a reviewer would receive.

01 RUN

```
python docs/demo/experiment.py --seed 0 --out run/
```

PATH	ROLE	
trials.csv · summary.csv · allocation_means.csv · contrasts.csv	aggregates	OFFICIAL
REPORT.md · run_config.json · figure.png	narrative, config, figure	OFFICIAL
private/ballots.csv	voter, choice	DENYLIST
scratch_notes.txt	lab leftover	UNMATCHED

A hash tool would digest the ballots. FarmNotary hashes nothing until a profile says what may leave.

The allowlist is the privacy model.

SHA256SUM, ON THIS FOLDER

These files match this digest.

Including `private/ballots.csv`. That is the entire claim.

FARM-NOTARY, PROFILE CONSENSUS

- Seven official artifacts. Policy recorded on the manifest.
- Ballots excluded even if a glob would admit them.
- `scratch_notes.txt` increments `unmatched_count`. The name is never printed.
- What you may claim. What you may not.

Nothing is hashed until you say what may leave.

01 PROFILE

```
farm-notary manifest --run-dir run/ --profile consensus --config run/run_config.json
```

ADMITTED	trials.csv, summary.csv, allocation_means.csv, contrasts.csv, REPORT.md, run_config.json, *.png
DENIED	any path containing ballot, vote, voter, individual_choice, or private
UNMATCHED	count only — scratch_notes.txt is not on the record

Profiles: consensus, rl-sweep, evolution-run. The resolved patterns are part of the claim.

Stamp the plan before the artifacts exist.

01 PLAN

```
farm-notary precommit --config run/run_config.json --command "python docs/demo/experiment.py --seed 0 --out '{run_dir}'"
```

02 BIND

```
farm-notary manifest --run-dir run/ --profile consensus --precommit precommit.json --config run/run_config.json --command "python docs/demo/experiment.py --seed 0 --out '{run_dir}'"
```

This room is dry-run. A lab would pass `--backend ots` on the plan, then run. FarmNotary will still stamp a favorable run after the fact. It will not call that a pre-specified design.

Reviewers read the card, not the exit code.

FARM-NOTARY VERIFY --RUN-DIR RUN/

DRY-RUN · THIS ROOM

```
level: none — no earned ladder level
next:  L0 — these bytes existed by time T; Bitcoin headers not verified by this tool (missing: Bitcoin attestation)
•  tamper-evident record          — pass
•  existed by time T              — missing
•  pre-specified design          — precommit bound
•  bitwise reproducible (scoped) — missing
•  not claimed: scientific correctness
```

Missing is not failure. Exit 0 means no attempted check failed — not that every claim was earned, and not that the science is right.

Change one official byte. The card breaks.

AFTER A ONE-LINE EDIT

Append a comment to `summary.csv`. Rehash fails. `tamper-evident record – fail`. The zip on Drive would still open.

AFTER RESTORE

Put the bytes back. The card returns to pass. That is the value versus a laptop folder: the official record is checkable without asking the author.

Immutability is not correctness. A manifest can perfectly notarize a wrong winner.

Same seed. New folder. Scope the sentence.

01 REPRODUCE

```
farm-notary reproduce --run-dir run/ --fresh-dir repro/
```

THIS TOY, SAME MACHINE

7/7

Every listed artifact matches. The only sentence the tool may emit:

byte-identical on x86-64 Linux in a pinned environment

NOT A CLAIM

- Cross-hardware bitwise identity.
- Linux ARM and macOS ARM receipts for the AgentFarm cell are not ok.
- Independently reproduced. This is the author's own re-run.

A byte-diff is not a science failure.

01 BREAK

```
python docs/demo/experiment.py --seed 0 --out broken/ --embed-path
```

WHAT CHANGED

`REPORT.md` baked the output directory. Re-run into a fresh folder changes one line. Science did not move. FarmNotary classifies `embedded_absolute_path`.

THE REAL CELL

AgentFarm consensus: 100 trials, 300 voters, 8 candidates, seed 0. The same bug took that record from 7/7 to 8/8 after `{run_dir}`. A classified byte-diff is **not a science failure**.

Opt in to execute. Do not cite Ln from the PDF.

01 READ

```
farm-notary verify --run-dir run/
```

02 OPT IN

```
farm-notary verify --run-dir run/ --verify-derived
```

03 SWEEP

```
farm-notary campaign --name demo-sweep --run-dir run/ --run-dir run-1/ --out sweep/
```

Step 02 only if you trust the recorded commands. `paper-pack` leaves reader ladder as `—`. A printed L0-L3 is not a finding.

The zip could not say this.

ADMISSION

Which files left. How many were omitted.
Ballots never hashed. Policy on the record.

LANGUAGE

A card that will not print a verified-result
badge. Missing is not failure. Last row is
always the non-claim.

RE-RUN

Same seed, scoped sentence. A path baked
into REPORT.md is classified, not treated as
a failed result.

An anchor proves existence, not correctness. Cross-hardware bitwise identity is not a claim.

Run the toy. Then hash a run you would cite.

01 NOTEBOOK

```
jupyter notebook docs/demo/farmnotary_live_demo.ipynb
```

02 INSTALL

```
pip install "farm-notary[ots]"
```

03 CITE

```
farm-notary anchor --run-dir path/to/run --backend ots --pin-remote pinata
```

github.com/Dooders/FarmNotary

Intro deck: docs/slides/index.html. Docs: PRINCIPLES · CLAIMS · DESIGN. Apache-2.0. Dry-run until you pass --backend ots.